



CurveMole User Manual

Preview 0.19.2

Sebastiano Romi

European Laboratory for Non-Linear Spectroscopy (LENS)

University of Florence (UNIFI)

`romi@lens.unifi.it`

Updated for CurveMole 0.19.2

GPL-3.0-or-later

Contents

1. About this manual	6
1.1 Intended audience	6
1.2 Version policy for the manual	6
1.3 Terminology and notation	6
1.4 Scientific responsibility	7
2. Core concepts	8
2.1 Project	8
2.2 Series and curves	8
2.3 Model, component, and parameter	8
2.4 Active, selected, and visible curves	8
2.5 Fit plan and result	9
3. Installation and first launch	10
3.1 Reference requirements	10
3.2 Desktop packages	10
3.3 Linux AppImage	10
3.4 Verifying SHA-256 checksums	11
3.5 Installing the Python package	11
3.6 Running from a source checkout	11
3.7 Confirming the installed version	11
3.8 Version badge and updates	12
4. A complete first fit	13
4.1 Import the example	13
4.2 Add a background	13
4.3 Add a Gaussian peak graphically	13
4.4 Adjust the model before fitting	13
4.5 Mask an interval if needed	14
4.6 Run the fit	14
4.7 Save and export	14
5. Main window and controls	15
5.1 Series and curves tree	15
5.2 Plot workspace	15
5.3 Navigation and axes	16
5.4 Model and parameters dock	16
5.5 Quick actions	17
5.6 Dockable tools	17
5.7 Themes	17
5.8 Drag and drop	18
6. Importing data	19
6.1 Supported text formats	19
6.2 Comments, blank lines, and encoding	19
6.3 Column mapping	19
6.4 Batch import	20
6.5 Invalid numeric cells	20
6.6 Uncertainties and weights	20
7. Curves, masks, and transformations	21
7.1 Immutable originals	21
7.2 Masking points and intervals	21

7.3 Mask targets	21
7.4 Undo and mask history	21
7.5 Data Calculator	22
7.6 Curve-to-curve calculations	22
7.7 Restoring original data	22
7.8 Worksheet	22
8. Building models	23
8.1 Component composition	23
8.2 Built-in peak parameterization	23
Gaussian	24
Lorentzian	24
Voigt	24
Pseudo-Voigt	24
8.3 Built-in backgrounds	24
Constant	24
Linear	25
Polynomial	25
Cubic spline	25
8.4 Quick Add Function	25
8.5 Adding a peak with the pointer	26
8.6 Adding a spline background with the pointer	26
8.7 Graphical editing after placement	26
8.8 Automatic peak suggestions	26
8.9 Duplicating, disabling, deleting, and reordering	27
8.10 Copying a fit to other curves	27
9. Parameters, constraints, and links	28
9.1 Parameter table	28
9.2 Parameter states	28
9.3 Parameter paths	28
9.4 Link evaluation	29
9.5 Safe expression language	29
10. Fitting	30
10.1 Pre-fit checklist	30
10.2 Fit modes	30
Single / independent	30
Sequential	30
Global simultaneous	31
10.3 Per-spectrum weights	31
10.4 Local least squares	31
10.5 Robust losses	31
10.6 Differential Evolution initial search	32
10.7 Cancellation and failed fits	32
10.8 Common causes of failure	32
11. Results, statistics, and diagnostics	33
11.1 Visual results	33
11.2 Reported global statistics	33
11.3 Per-curve statistics	33
11.4 Covariance and correlation	33
11.5 Residual diagnostics	34
12. Explicit uncertainty analyses	35
12.1 Parametric Monte Carlo	35
12.2 Residual bootstrap	35
12.3 Block bootstrap	35
12.4 Profile likelihood	35

12.5 Reproducibility and failed replicates	35
13. Projects, saving, and recovery	36
13.1 The .fitproj format	36
13.2 Atomic saves	36
13.3 Project locks and read-only mode	36
13.4 Normal save and portable copy	36
13.5 Autosave recovery	36
13.6 Unsaved changes	37
14. Exporting an analysis bundle	38
14.1 Starting an export	38
14.2 Default fit-results file	38
14.3 Optional outputs	38
14.4 Safe overwrite policy	39
14.5 Wide data tables	39
14.6 Tidy data and JSON	39
14.7 Reports and reusable files	39
14.8 Exporting one numeric file per spectrum	39
15. Command-line interface	41
15.1 General form	41
15.2 Open the GUI	41
15.3 List available functions	41
15.4 Fit one file	41
15.5 Fit a series	41
15.6 Inspect and validate	42
15.7 Export an existing project	42
15.8 Exit codes	42
16. Reproducible YAML workflows	43
16.1 Running a workflow	43
16.2 Complete example	43
16.3 Custom formulas in YAML	44
16.4 Trusted plugins	44
17. Python API	45
17.1 Stable package-level objects	45
17.2 Minimal fit	45
17.3 Bounds and fixed values	45
17.4 Cross-curve links	46
17.5 Global fit	46
17.6 API compatibility during Preview	46
18. Function Builder and plugins	47
18.1 Formula Builder	47
18.2 Custom function persistence	47
18.3 Local plugin structure	48
18.4 Trust boundary	48
18.5 Plugin Manager	48
19. Troubleshooting	49
19.1 The application does not start on Linux	49
19.2 Quick Guide, manual, updates, or issue links do not open	49
19.3 Windows SmartScreen or macOS Gatekeeper warns	49
19.4 Import produces one column	49
19.5 Imported points disappear from fitting	49
19.6 A peak appears at the wrong height after placement	49
19.7 A spline looks unstable	50
19.8 Differential Evolution requests finite bounds	50

19.9 The fit is underdetermined or covariance is unavailable	50
19.10 A parameter remains at a bound	50
19.11 A sequential fit pauses	50
19.12 A project opens read-only	50
19.13 Export refuses to overwrite	50
19.14 A resampling analysis has many failed replicates	51
19.15 An update is detected but cannot be installed automatically	51
19.16 Reporting a problem	51
20. Reproducible research, privacy, and citation	52
20.1 Recommended research record	52
20.2 Local processing and privacy	52
20.3 Citation	52
20.4 Author and contact	52
20.5 License	53
Appendix A. Keyboard shortcuts	54
Appendix B. Built-in identifiers	55
Appendix C. File extensions	56
Appendix D. Preview 0.19.2 limitations	57
Appendix E. Building this manual	58

CurveMole is a desktop-first, scriptable application for fitting one-dimensional scientific curves. This manual describes the behavior of CurveMole 0.19.2 Preview. The Markdown file is the authoritative source. The LaTeX source and PDF edition are generated automatically from it and must carry the same version number.

Preview status. CurveMole 0.19.2 is suitable for evaluation and controlled testing. It has not yet completed the scientific validation planned for version 1.0.0. Inspect the residuals, parameter correlations, constraints, and exported results before using a fit in research.

Useful links:

- [CurveMole repository](#)
- [Latest release and desktop downloads](#)
- [Issue tracker](#)
- [Zenodo record and DOI](#)
- [Quick Start](#)

1. About this manual

1.1 Intended audience

This manual is for researchers and technical users who need to fit spectra or other x-y data while retaining control over data preparation, model construction, constraints, uncertainty, and export. Typical data include:

- infrared and Raman spectra;
- powder diffraction patterns;
- kinetic traces;
- chromatographic profiles;
- temperature, pressure, or field-dependent signals;
- any other one-dimensional curve that can be represented by numeric x and y columns.

No programming knowledge is required for the graphical workflow. Separate chapters cover the command-line interface, YAML workflows, Python API, formula builder, and plugins.

1.2 Version policy for the manual

The version in the title and in the line at the top of this file identifies the CurveMole release whose behavior is documented. It is not an independent document version. For example, a manual marked 0.19.2 describes CurveMole 0.19.2.

The release automation performs four checks:

1. the application version and both manual version declarations must agree;
2. all local links in the Markdown source must resolve;
3. Pandoc must generate a standalone LaTeX document without warnings;
4. XeLaTeX must compile the generated source into a valid PDF.

The release assets use versioned names so that a downloaded manual can always be associated with the correct executable.

1.3 Terminology and notation

Menu commands are written as **File > Import data**. Buttons and field names are shown in **bold**. Commands, filenames, identifiers, and parameter paths use monospace text.

CurveMole uses the following residual convention in displays and exports:

$$r_i = y_i - f(x_i).$$

The optimizer minimizes the opposite sign, $f(x_i) - y_i$, after applying uncertainty and spectrum scaling. The sign has no effect on a least-squares minimum but matters when reading exported residuals.

1.4 Scientific responsibility

CurveMole can optimize a mathematically defined model, but it cannot decide whether that model is physically appropriate. A small residual or a high descriptive R^2 does not by itself establish uniqueness, chemical meaning, or predictive validity. In particular:

- do not add components solely to improve a scalar fit statistic;
- use bounds only when they have a defensible meaning;
- inspect correlations and active bounds;
- check whether residuals contain structure;
- report the exact CurveMole version, model, constraints, weighting, and uncertainty method used.

2. Core concepts

CurveMole keeps the scientific engine independent from the graphical interface. The desktop application, command-line interface, YAML workflows, and Python API all operate on the same domain objects.

2.1 Project

A project is the complete analysis workspace. It can contain data, models, masks, transformations, fit results, custom formulas, history, and interface state. Projects are saved as `.fitproj` files.

2.2 Series and curves

A **series** groups related curves. A **curve** contains one x array and one y array, with optional `sigma_x`, `sigma_y`, or point weights. Importing multiple y columns from a common file produces multiple curves in one imported series.

Original numeric arrays are stored as immutable 64-bit floating-point values. Data Calculator operations create reversible transformations instead of overwriting the original arrays.

2.3 Model, component, and parameter

Each curve has its own model. A model is an ordered sequence of components such as a background and several peaks. Every component has:

- a function, for example gaussian or linear;
- a user-facing name;
- a composition operator;
- a set of parameters;
- an enabled or disabled state;
- optional metadata, such as polynomial order or spline x nodes.

A parameter has a current value, optional lower and upper bounds, a fixed state, and an optional expression link. Standard errors and confidence limits are populated after a successful fit when they are available.

2.4 Active, selected, and visible curves

These states are deliberately distinct:

- **Active curve:** the single curve whose model and handles are being edited.
- **Selected curves:** one or more curves chosen for multi-curve operations.
- **Visible curves:** curves drawn in Overlay or Waterfall display modes.

Activating a curve does not automatically select every visible curve. Before a fit, mask transfer, or multi-curve calculation, check the target shown in the relevant dialog.

2.5 Fit plan and result

A fit plan specifies the curves, fit mode, solver settings, per-spectrum weights, and whether numerical contributions should be equalized. A fit result records the optimized parameters, statistics, residual arrays, covariance, correlation, warnings, timestamp, and solver status.

3. Installation and first launch

3.1 Reference requirements

The current reference configuration is:

Item	Minimum or supported target
Processor	x86-64, 4 CPU cores recommended
Memory	8 GB RAM
Storage	SSD recommended
Display	1920 x 1080 recommended
Windows	Windows 10 22H2 legacy compatibility or Windows 11, 64-bit
macOS	macOS 13 Ventura or newer, Intel or Apple Silicon
Linux	Ubuntu 22.04 LTS, Debian 12, or a comparable newer distribution
Python	Python 3.12 or newer for Python installations

A dedicated GPU is not required. Large multi-curve fits and resampling analyses may require more memory and time than the reference configuration.

3.2 Desktop packages

Open the [latest release](#) and select the package for your system.

Platform	Release asset	Normal launch method
Linux x86-64	CurveMole-VERSION-linux-x86_64.AppImage	Mark executable and run
Windows x86-64	CurveMole-VERSION-windows-x86_64.exe	Double-click the executable
macOS Apple Silicon	CurveMole-VERSION-macos-arm64.dmg	Open the DMG and launch the app
macOS Intel	CurveMole-VERSION-macos-x86_64.dmg	Open the DMG and launch the app

Windows and macOS packages are currently unsigned. The operating system may show a warning on first launch. Download only from the official GitHub release page and verify the checksum if the application will be used for research.

3.3 Linux AppImage

In a terminal opened in the download directory:

```
chmod +x CurveMole-0.19.2-linux-x86_64.AppImage
./CurveMole-0.19.2-linux-x86_64.AppImage
```

If the system cannot mount AppImages through FUSE, use extraction mode:

```
APPIMAGE_EXTRACT_AND_RUN=1 ./CurveMole-0.19.2-linux-x86_64.AppImage
```

Some minimal Linux installations may need graphical runtime libraries supplied by the distribution. On Debian or Ubuntu, the commonly required packages are:

```
sudo apt install libegl1 libgl1 libxkbcommon-x11-0 libxcb-cursor0
```

3.4 Verifying SHA-256 checksums

Every release includes SHA256SUMS.txt. On Linux:

```
sha256sum -c SHA256SUMS.txt --ignore-missing
```

On macOS:

```
shasum -a 256 CurveMole-0.19.2-macos-arm64.dmg
```

On Windows PowerShell:

```
Get-FileHash .\CurveMole-0.19.2-windows-x86_64.exe -Algorithm SHA256
```

Compare the reported value with the corresponding line in SHA256SUMS.txt.

3.5 Installing the Python package

Use an isolated environment. From a downloaded wheel:

```
python3 -m venv ~/.venv/curvemole
source ~/.venv/curvemole/bin/activate
python -m pip install ./curvemole-0.19.2-py3-none-any.whl
curvemole gui
```

On Windows PowerShell, activation is:

```
py -m venv $env:USERPROFILE\.venv\curvemole
& $env:USERPROFILE\.venv\curvemole\Scripts\Activate.ps1
python -m pip install .\curvemole-0.19.2-py3-none-any.whl
curvemole gui
```

3.6 Running from a source checkout

Clone the repository and install its locked development environment:

```
git clone https://github.com/SebRoLENS/curvemole.git
cd curvemole
uv sync --locked --group dev
uv run curvemole gui
```

For an editable installation using pip:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -e .
curvemole gui
```

3.7 Confirming the installed version

```
curvemole --version
```

The About dialog also displays the version. A project stores the application version that created it, and exports include the application version in machine-readable metadata.

3.8 Version badge and updates

The status bar contains a clickable version badge. CurveMole checks the latest GitHub release at startup and once per hour while the application remains open. The badge is:

- green when the installed release is current;
- yellow when a bug-fix update (x.x.y) is available;
- red when a feature (x.y.x) or major (y.x.x) update is available;
- grey while checking or when the check cannot be completed.

An automatic notification is shown only once for each newly detected release. Click the badge or choose **Help > Check for updates** to check again, inspect the release notes, or reopen the update dialog.

Linux AppImage and Windows standalone installations can normally use **Update now**. CurveMole downloads the matching release asset, verifies the SHA-256 digest supplied by GitHub, replaces the old executable, and removes obsolete release files. The application directory must be writable and no fit or uncertainty task may be running. Windows closes and restarts automatically; Linux asks the user to close and reopen the new AppImage. Python, source, macOS, unsupported architectures, and installations whose executable cannot be replaced must be updated manually from the release page.

4. A complete first fit

This tutorial uses `examples/gaussian.csv` from the source archive. It contains `x`, `y`, and `sigma_y` columns.

4.1 Import the example

1. Start CurveMole.
2. Choose **File > Import data** or press **Ctrl+I**.
3. Select `examples/gaussian.csv`.
4. In the mapping dialog, confirm comma delimiter, decimal point, and header.
5. Select `x` as **X column** and check `y` under **Y column(s)**.
6. Select `sigma_y` as the uncertainty type and column.
7. Press **OK**.

The curve appears in the left tree and becomes active. The central plot displays the data, and the model panel on the right identifies the active curve.

4.2 Add a background

1. Press the **+** button in **Model and parameters**, or press **Ctrl++**.
2. Select **Constant background**.
3. Keep composition set to add.
4. Press **OK**.
5. In the parameter table, enter an initial offset near the visual baseline.

4.3 Add a Gaussian peak graphically

1. Add another component and select **Gaussian**.
2. Press **OK**. The pointer changes to a crosshair.
3. Click at the approximate peak center.
4. Keep the mouse button pressed and drag horizontally from the center to either half-height edge.
5. Release the mouse button.

CurveMole uses twice the dragged distance as the initial FWHM. It estimates a positive peak height from the data minus the existing model, converts this height to the area parameterization, and adds the component as one undoable action.

If placement is incorrect, press **Ctrl+Z** and repeat. Press **Esc** during placement to cancel without adding the component.

4.4 Adjust the model before fitting

Select the Gaussian in the component list. Three green handles appear:

- drag the central target to change center and height;
- drag either vertical side handle to change FWHM;
- edit exact values or bounds in the parameter table.

For this example, a positive lower bound on sigma already exists intrinsically. You may enter a tighter scientifically motivated range, but it is not required.

4.5 Mask an interval if needed

Mask editing is active only while the graphical **Mask** toolbar button is enabled:

1. click **Mask**; the button becomes grey while masking is active;
2. in Single view it acts on the displayed spectrum; in Overlay/Waterfall choose the active spectrum or all visible spectra in the scope prompt;
3. right-drag to mask an interval, left-drag to restore it;
4. click **Mask** again to return to normal navigation.

The example does not normally require masking.

4.6 Run the fit

1. Press **F5** or choose **Fit > Fit**.
2. Use **Single / independent** mode.
3. Check that the example curve is enabled in the table.
4. Keep **Local constrained least squares** and linear loss.
5. Press **OK**.

After a successful fit, CurveMole commits the optimized parameter values back into the displayed model and immediately redraws the model sum, individual components, and residual panel. Parameter standard errors appear in the ± 1 sigma column when covariance is available. Open **View > Diagnostics** to inspect residual warnings.

4.7 Save and export

Save the complete workspace with **File > Save project as**. Use a .fitproj extension.

Then choose **File > Export analysis bundle**. Select a destination and choose both how to save and **What to export?**. The default selection writes only fit_results.csv, which associates every curve with its model functions and lists each parameter together with its fitted value and standard error. Data tables, plots, reports, JSON, reusable models, project copies, uncertainty matrices, and diagnostics are optional.

5. Main window and controls

5.1 Series and curves tree

The left dock contains a search field and a tree with **Visible**, **Series / Curve**, and **State** columns.

- Click a curve to activate it.
- Use Ctrl-click or Shift-click for a multi-selection, or use **Select all** and **Deselect all** above the tree.
- Use **Remove selected** to delete one or more accidentally imported curves. Removal is undoable.
- Use **New series** to create an empty named series. Empty series are retained when the project is saved.
- Right-click one or more selected spectra and choose **Move selected to series** to reorganize them.
- Right-click selected spectra and choose **Move selected up/down** to change their order inside a series. Multiple adjacent spectra move as a stable group.
- Right-click a series and choose **Merge series into** to append all its spectra to another series and remove the source series.
- Moving, merging, creating, and reordering series are all Undo/Redo operations and do not invalidate fitted parameters.
- Use the visibility checkbox to include or exclude it from Overlay and Waterfall displays.
- Double-click an editable name to rename a series or curve. Series names must remain unique.
- Right-click a **curve** and choose **Choose spectrum colour...** to set its colour. Red is reserved for the fitted Model sum and cannot be assigned to a spectrum.
- Right-click a **series** and choose **Series palette** to recolour the complete series with one of the built-in non-red palettes. Palette and individual colours are saved in the project.
- Use the search field to filter curves by name without removing them.

Curve states are:

State	Meaning
Not fitted	No successful fit is associated with the current curve state
Ready	The curve is prepared for fitting
Running	A fit involving the curve is in progress
Fitted	The last fit completed successfully
Modified/outdated	Data or model changed after the last successful fit
Failed	The latest fit attempt failed

5.2 Plot workspace

The central workspace contains the data/model plot and an optional residual plot. Controls above it provide:

- **Display:** Single, Overlay, or Waterfall;
- **X offset** and **Y offset:** display-only Waterfall spacing;
- **Residuals:** show or hide the linked residual panel.

The main toolbar contains the graphical **Mask** button. Its scope is chosen on entry in Overlay/Waterfall; mouse buttons determine the operation.

Offsets never modify data and are removed when converting pointer positions back to scientific coordinates. Fitting and export use the original numeric coordinates, not screen offsets.

The coordinate readout displays the pointer coordinate and the nearest finite point from the active curve. Overlay and Waterfall views use adaptive peak-preserving downsampling and clip drawing to the visible range for dense spectra. Single view keeps the full display resolution. Descending x arrays may be reversed only in the temporary display copy so that fast clipping remains available. Fitting, project storage, and export always use the complete original arrays.

Model functions receive systematic names such as **Voigt1**, **Voigt2**, and **Gaussian1**. Their labels are shown above each function maximum by default and are shifted vertically when necessary to avoid overlap. Right-click the main plot and toggle **Show component labels** to hide or show these labels. After every completed fit, CurveMole explicitly applies the returned optimized parameters to the displayed model, redraws and auto-ranges the plot so the newly fitted curves are visible immediately. The **Model sum is always red**; imported spectra and selectable series palettes exclude red so a data curve can never be confused with the fitted sum.

5.3 Navigation and axes

Normal plot navigation is available when neither placement nor Mask mode is active. The mouse wheel remains available during Quick Add peak placement, spline placement, Quick Fit, and live fit refreshes. A refresh preserves the current viewport instead of resetting a zoom chosen by the user. When the plot has keyboard focus, Up and Down activate the previous or next spectrum in project order, matching navigation in the curve tree.

Use **View > Axes** for:

- automatic range;
- logarithmic x or y;
- reversed x or y;
- locked plot view.

Reversing an axis is only a view setting. Logarithmic axes do not transform the data used by the fit. Values incompatible with a log display may not be visible even though they remain stored.

5.4 Model and parameters dock

The right dock normally lists components for the active curve. Enable **Show all functions** to list every model function in the project with its spectrum name. The list supports Ctrl-click and Shift-click multi-selection, including selections across spectra. Component order matters because composition operators are evaluated sequentially.

The controls below the list are:

Control	Action
+	Add a component
Duplicate	Duplicate the selected component and its settings
Up / Down	Change component order
Delete	Delete the component after confirmation
Copy fit	Copy selected model information to other curves

Control	Action
Copy parameter	Copy one parameter from one source function to chosen project-wide targets

Bulk actions on selected functions can delete, duplicate, reorder, enable or disable, mark or unmark backgrounds, and fix or free their parameters as applicable. Each bulk edit is one Undo/Redo operation. Uncheck a component in the list to disable it without deleting it. Disabled components are not evaluated or fitted.

Copy parameter is enabled only when exactly one source function is selected. Select a row in the parameter table to make that parameter the default, open the copy window, and explicitly check the target functions. The window shows the source spectrum, function, parameter, and numeric value. By default only the value is copied; optional controls also copy bounds, fixed/free state, and the parameter link/relation. Targets without the parameter or whose retained bounds reject the source value are skipped and reported. The complete copy is undoable.

5.5 Quick actions

The main toolbar includes **Quick Add Function** and an adjacent function selector. The selector contains the complete current registry: built-ins, project formulas, reusable user formulas, and loaded plugin functions. CurveMole remembers the last selection across sessions and also updates it when a function is added through the ordinary Add dialog.

For peak functions, Quick Add starts click-drag graphical placement. For a cubic spline it starts node placement. Other functions are inserted immediately with their default parameter values. Quick Add remains available repeatedly until the user changes the selected function or tool.

The toolbar also provides the most common project, model, fitting, background, and calculator actions as icons. Hover over an icon to see its action name.

5.6 Dockable tools

The **View** and **Tools** menus show or hide these docks:

- **Worksheet:** read-only view of current x, y, uncertainty, weight, and mask state;
- **Diagnostics:** residual summary and warnings;
- **Log:** operational messages and errors;
- **Data Calculator:** reversible scalar and curve-to-curve transformations;
- **Function Builder:** safe mathematical custom functions;
- **Uncertainty Analysis:** Monte Carlo, bootstrap, and profile calculations.

Docks can be moved, tabbed, resized, or floated. CurveMole remembers window geometry, dock arrangement, and theme. Use **View > Reset layout** to restore the standard layout.

5.7 Themes

View > Theme provides System, Light, and Dark themes. Theme choice affects only presentation. It does not change plot data, exported numeric values, or fitting.

5.8 Drag and drop

Drop a .fitproj file onto the window to open it. Drop any other local file to begin content-aware data import, regardless of its extension. If a drop contains both a project and data files, the first project is opened.

6. Importing data

6.1 Supported text formats

The graphical importer shows .txt, .dat, .csv, and .tsv as common choices but can open any text-based file whose contents form a valid numeric table. For example, spectrometer files named .xy, .xye, or with a vendor-specific suffix can be selected through **All files** or dropped onto the window. The extension is not used as proof of validity. Files must contain at least two columns and at least two rows after parsing.

CurveMole detects common delimiters:

- whitespace;
- comma;
- semicolon;
- tab;
- vertical bar.

It also proposes decimal point or decimal comma and whether the first data row is a header. Always inspect the preview before accepting the mapping.

6.2 Comments, blank lines, and encoding

Blank lines are ignored. Lines whose first non-space character is #, ;, %, or ! are treated as comments by the default importer. UTF-8 with an optional byte order mark is the default encoding.

CurveMole also detects leading instrument metadata or preamble lines before the rectangular table. **Ignore first rows** shows the proposed number of physical lines to skip and can be changed from 0 to 1,000,000. Skipping occurs before delimiter, decimal, and header interpretation; therefore a column-name row after the preamble can still be used as the first data-row header. The accepted value is stored in the curve import metadata for reproducibility.

The semicolon can therefore be ambiguous: a line beginning with semicolon is a comment, while semicolons inside table rows can be delimiters. Check the preview for locale-specific exports.

6.3 Column mapping

Choose exactly one x column and one or more y columns. Each selected y column becomes a separate curve sharing the selected x column. Use **Select all Y columns** or **Deselect all Y columns** when importing tables with many signals. The Python API additionally supports explicit x-y column pairs.

Optional columns are:

- sigma_x;
- sigma_y;
- generic weight;
- variance;
- inverse variance.

Only one of `sigma_y`, generic weight, variance, or inverse variance can be selected for a given import. Variance is converted to `sigma_y` by square root. `sigma_x` is stored in the project but is not used by the version 0.19.2 optimizer.

6.4 Batch import

When importing multiple files, **Apply this mapping to all files in this batch** reuses the first file's parsing and column mapping. Enable it only if all files truly share a layout. CurveMole does not silently guess a new mapping for an incompatible file in the same batch.

6.5 Invalid numeric cells

Non-numeric cells are converted to missing numeric values. Rows are retained in the stored arrays but excluded from fitting when `x` or `y` is not finite. A row is also excluded if:

- `sigma_y` is non-finite or not strictly positive;
- a point weight is non-finite or negative;
- it is masked;
- it lies outside active fit ranges defined through the core API.

CurveMole does not reorder rows, average duplicate `x` values, or interpolate missing values during import.

6.6 Uncertainties and weights

If absolute `sigma_y` values are supplied, point residuals are scaled as:

$$r_i^{(w)} = \frac{f(x_i) - y_i}{\sigma_{y,i}}.$$

If inverse variances $w_i = 1/\sigma_i^2$ are supplied, CurveMole multiplies residuals by $\sqrt{w_i}$. A generic weight column directly multiplies residuals by its value. This distinction is intentional and is recorded in the curve metadata.

Zero generic weight is allowed and gives a point zero numerical influence. A zero or negative `sigma_y` is invalid and excludes the row.

7. Curves, masks, and transformations

7.1 Immutable originals

CurveMole preserves imported x, y, uncertainty, and weight arrays. Displayed working arrays are reconstructed by applying a transformation history to the originals. This design makes data preparation reversible and auditable.

7.2 Masking points and intervals

Masked points remain stored but are excluded from fitting. They are drawn as faded markers, and masked intervals are shaded.

Enable the graphical **Mask** button: right-drag masks an interval and left-drag unmasks it. A right/left click performs the equivalent operation on a point. The button becomes grey while active. Turn it off to navigate without changing exclusions. Changing display mode also exits masking.

7.3 Mask targets

Target	Effect
Active	Changes only the active curve
All visible	Changes every currently visible curve

Single view always uses the active spectrum. Overlay and Waterfall prompt for Active or All visible when entering Mask mode; cancelling leaves masking off.

For interval masks, the same numeric x interval is applied to every target. For a single-point mask transferred from the active curve, CurveMole locates the nearest x value on other targets and applies it only if it lies within the configured tolerance.

Set this tolerance through **Data > Mask transfer tolerance**. A value of zero requires an exact matching x value on transferred point masks.

7.4 Undo and mask history

A graphical mask action is placed on the application Undo stack. Press **Ctrl+Z** to undo it or **Ctrl+Shift+Z** on platforms that use that standard Redo binding. The project also stores mask arrays and numeric mask intervals.

Fits, including Quick Fit and sequential fitting, also support Undo/Redo. Undo restores the pre-fit models, parameter values, fit results, and sequential pause state; Redo restores the completed fit without running the solver again. Cancelled or failed worker operations restore the previous state. The Undo stack retains the latest 20 operations (fits and other edits combined); input data arrays are not duplicated for fit history. Undo/Redo are disabled while a worker is running.

Quick Fit is available before opening Fit...: its first run uses the default independent-fit settings on the selected curves, falling back to the active curve. Later runs reuse the last configured fit settings. Open Fit... to adjust the solver or fitting mode.

7.5 Data Calculator

Open **Tools > Data Calculator**. The following scalar operations are available:

- add to or subtract from y;
- multiply or divide y;
- shift x;
- scale x;
- normalize y by maximum absolute value;
- normalize y by signed integrated area.

Choose Active curve, Selected curves, or Entire series as the target. Normalization also scales `sigma_y` consistently. Dividing by zero and normalizing a zero or invalid signal are rejected.

7.6 Curve-to-curve calculations

The calculator can add, subtract, multiply, or divide by another curve. The operand is aligned to the target x grid by:

- linear interpolation;
- nearest-neighbor interpolation;
- cubic spline interpolation.

Extrapolation is disabled by default. Outside the operand range, aligned values become missing and will therefore be excluded from later fitting. Enabling extrapolation is an explicit advanced choice. Cubic interpolation requires at least four unique finite operand x values.

The aligned operand array is stored with the transformation so that reopening the project does not depend on a later change to the source curve.

7.7 Restoring original data

Restore original data removes all currently applied transformations from the selected target while retaining them on the redo stack. The imported arrays are not re-read from disk. The whole action is also integrated with application Undo.

7.8 Worksheet

The Worksheet is a read-only inspection view. It shows up to 100,000 rows for the active curve with columns for x, y, `sigma_y`, weight, and mask state. It is not a spreadsheet editor and does not modify source arrays.

8. Building models

All entries in the function library are fitting functions. Constant, linear, polynomial, cubic-spline, peak-shaped, and custom formulas are not separated into a special background-function class. Background is instead a property of a model component. Select a component in **Model and parameters** and enable **Mark as background** when that component represents the experimental baseline or background.

Data > Subtract background... uses these component-level marks. If no component is marked, CurveMole first asks which model functions should be designated as background. If background components already exist, it asks which of the marked components should be subtracted. The subtraction uses the current resolved/fitted parameter values, is reversible with Undo, applies over the complete data array, and disables the subtracted components afterwards to prevent double-counting.

The same dialog can apply the operation to every eligible spectrum, using the enabled functions already marked as background in each model. **Data > Revert background...** lists only spectra changed by this dedicated subtraction command and restores the selected ones, including the relevant function states. **Visual only - background-subtracted** instead removes the enabled marked background only from the plot; it never changes curve arrays, fit inputs, or exported values.

During graphical cubic-spline placement, nodes may be placed anywhere in plot coordinates, including outside the x/y extent of the measured data. Adding nodes does not auto-range the graph. Left-drag continues to pan and the mouse wheel continues to zoom while placement is active.

8.1 Component composition

Components are evaluated in list order. Available operators are:

Operator	Operation
add	Add the component array to the accumulated model
subtract	Subtract the component array
multiply	Multiply the accumulated model by the component array
divide	Divide the accumulated model by the component array
convolve	Convolve using FFT and scale by median x spacing

The first enabled component can use only add or subtract; multiplication, division, or convolution requires an existing accumulated model. Additive peak plus background models should normally use add for every component.

Division by a component that reaches zero and other non-finite model evaluations are rejected by the fitter.

8.2 Built-in peak parameterization

Every built-in peak uses signed integrated area, not peak height, as its intensity parameter. Negative areas produce negative peaks. This common parameterization makes areas comparable between different line shapes.

Gaussian

Parameters: area, center, sigma.

$$G(x) = \frac{A}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - x_0}{\sigma} \right)^2 \right].$$

The derived FWHM is 2.354820045σ . sigma is intrinsically positive.

Lorentzian

Parameters: area, center, gamma.

$$L(x) = \frac{A\gamma}{\pi[(x - x_0)^2 + \gamma^2]}.$$

gamma is the half width at half maximum, so FWHM is 2γ . gamma is intrinsically positive.

Voigt

Parameters: area, center, sigma, gamma.

The Voigt component is the area-normalized convolution of a Gaussian with standard deviation sigma and a Lorentzian with HWHM gamma. CurveMole evaluates it with the SciPy Voigt profile. Both width parameters are intrinsically positive.

The displayed FWHM is the approximation:

$$\text{FWHM} \approx 0.5346(2\gamma) + \sqrt{0.2166(2\gamma)^2 + (2.354820045\sigma)^2}.$$

Pseudo-Voigt

Parameters: area, center, fwhm, eta.

The profile is an area-normalized linear mixture with a common FWHM:

$$pV(x) = A[(1 - \eta)G_{\text{unit}}(x) + \eta L_{\text{unit}}(x)].$$

fwhm is intrinsically positive and eta is intrinsically bounded to $[0, 1]$. eta=0 is Gaussian and eta=1 is Lorentzian.

8.3 Built-in backgrounds**Constant**

Parameter: offset.

$$B(x) = c.$$

Linear

Parameters: intercept, slope.

$$B(x) = c_0 + c_1x.$$

Polynomial

Choose an order from 0 to 50 when adding the component. Parameters are c_0 , c_1 , and so on through the chosen order:

$$B(x) = \sum_{k=0}^n c_k x^k.$$

High polynomial orders are numerically sensitive when x values are large or poorly scaled. Prefer a low order or rescale x with the Data Calculator when scientifically appropriate.

Cubic spline

Spline x nodes are fixed metadata. Their y values (y_0 , y_1 , and so on) are normal parameters that may be fitted, fixed, bounded, or linked. Two nodes produce linear interpolation. Three or more nodes produce a natural cubic spline, with extrapolation outside the node range.

New graphically placed spline y nodes are fixed by default so an intentionally drawn baseline does not move when fitting begins. Uncheck **Fixed** for individual nodes, or use **Lock all** / **Unlock all** below the parameter table for every parameter in the selected function. The spline remains continuous through masked intervals even though masked measurements do not contribute to the objective function.

8.4 Quick Add Function

Choose a registered function from the selector beside **Quick Add Function** in the main toolbar. The list includes built-in peaks and backgrounds, generic functions, project-contained formulas, reusable formulas, and trusted plugin functions. The last choice is remembered.

- a peak opens graphical centre/FWHM placement;
- a cubic spline opens point-by-point node placement;
- every other function is added immediately with its default parameters.

Quick Add therefore uses the same registry and initialization logic as the normal Add dialog. Selecting a function in the normal Add dialog also makes it the next Quick Add choice.

8.5 Adding a peak with the pointer

After choosing a peak and pressing **OK**:

1. click the intended center;
2. drag horizontally to either half-height edge;
3. release to set the initial full width.

A green region previews the width. If no horizontal width is selected, CurveMole uses a default based on 5 percent of the x span, local point spacing, and floating point precision.

For built-in peaks, CurveMole converts the graphical height and FWHM into the native area and width parameters. New custom peaks should declare explicit semantic roles for their position, intensity, and width parameters in Function Builder. Older custom functions without role metadata retain compatibility through conventional names such as center, x0, fwhm, sigma, gamma, area, amplitude, or height.

8.6 Adding a spline background with the pointer

After choosing **Cubic-spline background**:

1. left-click background nodes on the plot;
2. inspect the dashed preview, which updates after each click;
3. use **Undo point** if necessary;
4. finish with **Finish** or a right-click after at least two nodes;
5. press **Esc** to cancel the entire placement.

Nodes are sorted by x. Clicking again at effectively the same x updates that node instead of creating a duplicate. Exact duplicate x positions are not permitted.

8.7 Graphical editing after placement

Select a peak component to display its handles. Drag the central target to change center and area-derived height. Drag a side line to change FWHM. For a Voigt peak, changing FWHM graphically scales sigma and gamma by the same ratio, preserving their current relative contribution.

Select a cubic spline to drag its y nodes. Node x positions remain fixed. Exact numeric edits are available in the parameter table.

Fixed values cannot normally be changed by a drag. Hold **Ctrl** while dragging to change the stored value while leaving the parameter fixed. Linked values cannot be dragged because their value is controlled by an expression.

8.8 Automatic peak suggestions

Choose **Model > Find Peaks**. The first dialog can search positive, negative, or both signs. The next dialog selects any registered function classified as a peak, initially preferring the current Quick Add peak. CurveMole then:

- subtracts the median as an initial baseline;
- estimates noise through a robust median absolute deviation;
- uses a default prominence of the larger of three noise estimates or 1 percent of the y range;
- estimates width at half prominence;
- sorts suggestions by prominence.

Choose how many suggestions to add. Every suggestion is initialized using the selected peak function's explicit parameter roles or, for legacy functions, recognized parameter names. Suggestions are initial estimates, not a scientific decision about the number, identity, or line shape of peaks.

8.9 Duplicating, disabling, deleting, and reordering

Duplicate a component to create an adjacent copy with a new internal identifier. Disable a component to compare models without losing its settings. Deletion and reordering are undoable. Reordering can materially change models that use operations other than addition or subtraction.

8.10 Copying a fit to other curves

Model > Copy fit copies selected information from the active curve. Use **Select all** or **Deselect all** when choosing many target curves. Options include:

- component structure;
- current or best values;
- bounds and fixed states;
- internal links;
- background components;
- masks;
- fit ranges.

Masks and fit ranges are excluded by default because x grids and excluded regions may not correspond scientifically. When copying masks, point transfer uses the configured x tolerance. Review every target model after copying.

9. Parameters, constraints, and links

9.1 Parameter table

The table columns are:

Column	Meaning
Parameter	Parameter name; linked parameters show a link indicator
Value	Current value or successful fitted value
± 1 sigma	Covariance-based standard error when available
Fixed	Exclude the parameter from optimization
Lower	Lower bound; blank means negative infinity
Upper	Upper bound; blank means positive infinity
Link	Graphical Set link... control; linked parameters show their source in readable form

Press **Set link...** to choose a source spectrum, component, and parameter. The default relationship is **Equal to source**. Choose **Advanced expression** for relationships such as $2 * \text{\texttt{\$ \{source\}}} + 1$. Use **Remove link** to make the parameter independent again.

An edit is validated immediately. CurveMole rejects a value outside its bounds, a lower bound above an upper bound, an invalid expression, a missing referenced parameter, a link cycle, or a linked value that violates its own bounds.

9.2 Parameter states

A parameter is exactly one of:

- free;
- fixed;
- lower-bounded;
- upper-bounded;
- interval-bounded;
- linked.

Intrinsic function bounds always remain active. For example, clearing the visible lower cell cannot make a Gaussian sigma negative because the function definition sets a positive intrinsic minimum.

9.3 Parameter paths

The canonical path is:

`curve_id.component_id.parameter_name`

A link refers to a path inside `\{\dots\}`. For example:

```
${curve_ab12.component_cd34.center}
```

or a derived relation:

```
${curve_ab12.component_cd34.center} + 12.5
```

The graphical **Set link...** dialog creates these paths automatically, so normal GUI use does not require knowing curve or component identifiers. The canonical expression is still shown as a tooltip and remains available to advanced workflows, Python code, and machine-readable exports. For a link between different spectra, select both spectra and use **Global simultaneous** mode; CurveMole blocks incompatible fit modes with an explicit message.

9.4 Link evaluation

Links form a directed dependency graph. CurveMole resolves the graph before fitting, detects cycles, and evaluates linked values at each model evaluation. Linked parameters are not independent optimizer variables. If covariance is available, their standard errors are propagated numerically from free parameters.

9.5 Safe expression language

Formulas and links are parsed as a restricted expression tree. They are never passed to unrestricted Python `eval` or `exec`.

Allowed arithmetic includes `+`, `-`, `*`, `/`, power `**`, remainder `%`, unary signs, comparisons, boolean combinations, and conditional expressions. Constants are `pi`, `e`, and `inf`.

Allowed functions are:

```
abs, sqrt, exp, log, log10,  
sin, cos, tan, arcsin, arccos, arctan,  
sinh, cosh, tanh,  
erf, erfc,  
minimum, maximum, clip, where, heaviside
```

Attribute access, imports, indexing, comprehensions, arbitrary function calls, file access, and network access are not part of the expression language.

10. Fitting

10.1 Pre-fit checklist

Before pressing **F5**:

1. confirm the active and selected curves; use **Select all** / **Deselect all** when appropriate;
2. inspect invalid and masked points;
3. choose a physically defensible background and peak count;
4. provide reasonable initial values;
5. add justified bounds for poorly determined parameters;
6. inspect links for cycles or unintended dependencies;
7. check that each selected model contains at least one free parameter.

10.2 Fit modes

Single / independent

Each selected curve is fitted as an independent problem. Results are then combined for reporting. Parameter links between different independently fitted curves cannot act as a simultaneous cross-curve constraint; use Global mode for that purpose.

Sequential

Sequential mode is a propagating refinement. Choose an **Initial source spectrum** whose model and parameters have already been inspected and approved. The source is not re-fitted. Its model is cloned to the next selected spectrum, fitted there, and the result becomes the source for the following spectrum. Selected spectra before the chosen source are excluded; target spectra do not need models in advance.

Use **Functions excluded from copying** to select any source functions to leave out. Functions already present on each target retain their values, metadata, constraints, and order during copying; the fit still respects their fixed/free state. A function with the same internal identity is kept rather than duplicated. Distinct functions remain distinct even if their names match. Target-local functions are not carried into later spectra. The selection is also preserved when resuming a paused sequence. This is independent of ignoring functions in the parameter-change pause monitor.

For newly copied functions, parameter values and structure are propagated. The dialog separately controls whether to preserve bounds, fixed/free state, internal links, background tags, enabled/disabled state, and composition/grouping. Internal links are remapped to corresponding target parameters. Structural metadata required by the function, such as spline nodes and custom-formula metadata, is always retained.

Two optional safeguards can pause a sequence:

- normalized residual RMSE worsens by both a selected factor and a minimum fraction of the signal scale; defaults are 2.5 times and 2 percent;
- a free parameter changes by more than a selected normalized amount; the default is 75 percent.

Specific source functions can be excluded from the parameter-change trigger while remaining fully copied, fitted, and included in the residual. If fitting or a safeguard pauses the sequence, the affected spectrum becomes active and a persistent **Continue sequential fit** button appears in the status bar. Inspect or edit that spectrum, then continue; the accepted current model becomes the new propagation source. CurveMole does not silently skip a paused spectrum.

Global simultaneous

All selected residual arrays are concatenated into one optimization problem. Models may differ between curves. Parameter links can connect curves and express shared, offset, scaled, or otherwise related values.

10.3 Per-spectrum weights

The Fit dialog accepts a positive numeric spectrum weight for every selected curve. If s_j is the spectrum weight, all point residuals from spectrum j are multiplied by $\sqrt{s_j}$.

Scale each spectrum to equal numerical contribution additionally divides the residual vector of each curve by the square root of its number of usable points. This prevents a densely sampled curve from dominating only because it contains more rows. It does not normalize signal amplitude or uncertainty and is never enabled silently.

10.4 Local least squares

The default is local nonlinear least squares using SciPy. CurveMole automatically selects:

- Levenberg-Marquardt for an unbounded ordinary least-squares problem;
- trust-region reflective when bounds or a robust loss require it.

The graphical dialog exposes the maximum number of evaluations and confidence level. New fit plans default to 1,000 evaluations and use a step tolerance that permits a successful early stop when parameter changes become numerically negligible. During local optimization, the status progress is updated and the plotted trial model is refreshed every 20 evaluations. These refreshes preserve the user's current zoom and do not replace the stored parameters unless the fit completes successfully. The Python API and YAML settings additionally expose tolerances, scaling, local method, random seed, and Differential Evolution controls.

10.5 Robust losses

Available losses are:

- linear for ordinary least squares;
- soft_ll;
- huber;
- cauchy.

Robust loss can reduce the influence of large residuals, but it is not a substitute for understanding outliers, detector artifacts, or an incomplete model. AIC, AICc, and BIC are reported only for the linear loss. Robust-loss covariance uses a sandwich estimate, and CurveMole recommends explicit resampling.

10.6 Differential Evolution initial search

Choose **Differential Evolution + local refinement** when a local initial estimate is not sufficient. Differential Evolution searches globally, then passes its best point to the local least-squares solver.

Every free parameter must have finite user bounds. CurveMole will not invent search bounds because arbitrary limits can change scientific conclusions. Global search is usually much slower than a well-initialized local fit.

10.7 Cancellation and failed fits

Use **Fit > Cancel running task** to request cancellation. The request is checked during residual evaluation and resampling. The window cannot close while a background task remains active.

If the solver raises an error or does not converge, CurveMole restores the last valid parameter values instead of committing an incomplete optimizer vector. The curve is marked Failed, the error is written to the Log, and the last successful result remains separate from the failed attempt.

10.8 Common causes of failure

- no free parameters;
- fewer than two usable points;
- initial value outside bounds;
- non-finite model values;
- division by zero in a composed model;
- a missing or cyclic link;
- a linked value outside its bounds;
- unbounded parameters with Differential Evolution;
- an underdetermined model;
- inadequate initial values or maximum evaluations.

11. Results, statistics, and diagnostics

11.1 Visual results

The main plot shows data, total model, individual component curves, and residuals. The parameter table shows fitted values and standard errors. A curve state changes to Fitted only after solver success.

The full numeric result is stored in the project and written to exports. Preview 0.19.2 does not yet provide a single comprehensive on-screen results table, so use the analysis bundle for archival inspection.

11.2 Reported global statistics

Let N be the number of usable observations and k the number of independent free parameters. CurveMole reports:

Statistic	Definition or interpretation
Degrees of freedom	$N - k$
RSS	$\sum_i (y_i - f_i)^2$ without point weighting
RMSE	$\sqrt{\text{RSS}/N}$
Chi-square	Sum of squared weighted residuals
Reduced chi-square	Chi-square divided by $N - k$ when positive
Descriptive R^2	$1 - \text{RSS}/\text{TSS}$
AIC	Gaussian ordinary least-squares expression when applicable
AICc	Small-sample correction when $N > k + 1$
BIC	Gaussian ordinary least-squares expression when applicable

The descriptive R^2 can be misleading for baselines, constrained fits, or models without an intercept. It is included as a familiar descriptor, not a universal model quality score.

11.3 Per-curve statistics

Each curve output records the original usable indices, x , observed y , fitted y , residual, weighted residual, N , RSS, RMSE, and descriptive R^2 . The global result combines all curves in the plan.

11.4 Covariance and correlation

For ordinary least squares, covariance is derived from the fit Jacobian. If all curves provide `sigma_y`, the uncertainty is treated as absolute by default and the covariance is not rescaled by residual variance. Otherwise residual variance scales the covariance.

CurveMole uses a pseudo-inverse when the Jacobian is rank deficient and reports a warning. Covariance is unavailable when degrees of freedom are not positive. A warning is also generated when at least one free-parameter correlation has absolute value 0.95 or greater.

Symmetric normal intervals are clipped to active bounds. A parameter at a bound is flagged because its true uncertainty is generally asymmetric and not adequately summarized by a symmetric covariance interval.

11.5 Residual diagnostics

Open **View > Diagnostics** after a fit. The summary reports:

- Durbin-Watson statistic;
- number of residuals with absolute standardized value at least 3;
- lag-1 autocorrelation;
- warnings for potential outliers and autocorrelation beyond an approximate 95 percent white-noise band.

The analysis bundle exports the autocorrelation sequence for every fitted curve. A structured residual pattern usually indicates that the model, background, weighting, or data treatment deserves review.

12. Explicit uncertainty analyses

Open **Fit > Uncertainty Analysis** only after a successful fit. These calculations run in the background, can be cancelled, and may be substantially slower than the baseline fit.

12.1 Parametric Monte Carlo

This method generates synthetic y values from the fitted model plus independent normal noise using the imported absolute sigma_y array. Every selected curve must therefore provide valid sigma_y.

Each synthetic data set is refitted using the local solver. The result records the requested, completed, and failed replicates, seed, free parameter paths, samples, empirical intervals, and failure messages.

12.2 Residual bootstrap

Residuals are centered, sampled independently with replacement, added to the fitted model, and refitted. This assumes residual exchangeability and can be inappropriate when residuals are serially correlated or heteroscedastic.

12.3 Block bootstrap

The block bootstrap resamples contiguous circular residual blocks. Enter a block length or choose Automatic. The automatic estimate uses the first lag at which the absolute residual autocorrelation falls below e^{-1} , subject to practical limits.

Block resampling is often preferable for spectra or kinetic traces with local correlation, but the block length remains a scientific modeling decision.

12.4 Profile likelihood

Choose one independent, non-linked parameter. CurveMole fixes it across a finite grid, refits all remaining free parameters, and compares chi-square with the baseline. The default grid spans approximately three covariance standard errors on either side, or a fallback span when no standard error is available, while respecting bounds.

Preview 0.19.2 uses 31 grid points and a one-parameter chi-square threshold. Failed grid points are counted. A profile interval is more informative than a symmetric standard error near bounds or in nonlinear problems, but grid resolution should be considered when interpreting endpoints.

12.5 Reproducibility and failed replicates

The default seed is 1729 unless changed through programmatic settings. Resampling outputs record configuration and up to the first 100 failure messages. Do not report an empirical interval without also reporting how many replicates completed and failed.

13. Projects, saving, and recovery

13.1 The .fitproj format

A .fitproj is a ZIP container with:

- a versioned JSON manifest;
- original arrays stored as NumPy .npy payloads;
- uncertainty and weight arrays when present;
- mask arrays;
- transformation operands;
- models, links, custom functions, results, history, and UI/export state;
- SHA-256 checksums for every binary payload.

NumPy arrays are loaded with `allow_pickle=False`. CurveMole projects never require executing a Python pickle. The format is documented in [project-format.md](#).

13.2 Atomic saves

CurveMole writes a temporary sibling file, validates the archive, and then atomically replaces the destination. A failed save should therefore not partially overwrite a previous valid project.

13.3 Project locks and read-only mode

Opening a project creates a sibling .lock file containing process and host information. If another instance already holds the lock, CurveMole opens the project read-only. Use **Save project as** to create an editable copy. A stale lock left by an abnormal termination may need manual inspection and removal after confirming that no other instance is using the project.

13.4 Normal save and portable copy

Save project updates the current project path. **Save portable copy** writes a complete separate .fitproj without changing the active project path. Since project data are already embedded, the portable copy is intended for sharing or archiving a snapshot.

Projects are not encrypted. Use suitable filesystem permissions or encrypted storage for confidential data.

13.5 Autosave recovery

Every ten minutes, a modified revision is written to the operating system's CurveMole user cache. An unchanged revision does not create another recovery. The three newest valid, distinct recovery files for a project are retained.

Recovery files end in .fitproj and can be opened through the normal Open Project dialog if needed. Saving the project normally clears its recovery files. Preview 0.19.2 does not yet show an automatic recovery chooser at startup, so after an abnormal termination inspect the CurveMole user cache before clearing application data.

13.6 Unsaved changes

An asterisk in the window title marks a dirty project. New Project, Open Project, and Quit ask whether to save, discard, or cancel when unsaved changes exist.

14. Exporting an analysis bundle

14.1 Starting an export

Choose **File > Export analysis bundle** or press **Ctrl+E**. Export mode and export content are independent.

The existing save modes remain:

- leave both mode checkboxes clear to create a new export in the selected directory;
- **Create versioned export** to create a timestamped subdirectory;
- **Update existing CurveMole-owned files after confirmation** to update files that CurveMole can verify as belonging to the current project export.

Under **What to export?**, select the desired outputs. Only **Fit results** is checked by default.

14.2 Default fit-results file

The default export creates only `fit_results.csv`. No additional visible file and no empty subdirectory is created.

Each row represents one parameter of one model component and includes the series, curve and curve ID, component and component ID, displayed function and function ID, enabled/background state, composition operator, parameter name and full parameter path, fitted value, standard error, confidence interval, bounds, fixed/link state, unit, human-readable value with uncertainty, and derived area/FWHM where defined.

This format makes the association between every data curve, the functions used to fit it, and the fitted parameters explicit in a single table.

14.3 Optional outputs

The dialog can additionally export:

- data plus fitted-function wide CSV tables;
- a tidy CSV table for Python;
- machine-readable fit results as JSON;
- a reusable `.fitmodel`;
- a `.fitproj` project copy, optionally including full uncertainty samples;
- the main plot as PNG and/or SVG;
- compact HTML and PDF summaries;
- a full reproducibility HTML report;
- covariance and correlation matrices when available;
- residual autocorrelation diagnostics when available;
- an export README.

Directories such as `data/`, `python/`, `figures/`, `report/`, `uncertainty/`, and `diagnostics/` are created only when a selected output actually writes a file there. An option whose required fit result is unavailable does not create an empty folder.

14.4 Safe overwrite policy

New exports store their ownership list in the project export configuration rather than creating a hidden sidecar file. This allows the default export to contain truly only `fit_results.csv`. Existing `.curvemole-export.json` manifests from older versions are still read for backwards-compatible updates.

When **Update existing** is selected, CurveMole refuses to overwrite a colliding file unless it can verify that the file belongs to the current project export. Unrelated files in the directory are preserved.

14.5 Wide data tables

Optional wide CSV files are intended for direct inspection and tools such as Origin or QtiPlot. Depending on data and model, columns include:

- x and y;
- total fit;
- residual as data minus fit;
- total background;
- every enabled component;
- `sigma_y` or point weight;
- mask flag.

Rows remain aligned with the complete curve, including masked and invalid entries.

14.6 Tidy data and JSON

`python/data_tidy.csv` stores one quantity per row with series, curve, component, x, value, mask, and validity fields. `python/results.json` records schema version, application version, result metadata, parameters, statistics, warnings, and settings without duplicating large numeric arrays.

14.7 Reports and reusable files

- `summary_report.html` is a compact human-readable report.
- `report/full_reproducibility.html` includes project metadata.
- `report/summary.pdf` contains a summary plot and statistics.
- `.fitmodel` stores the first project model and custom formulas for programmatic reuse.
- `.fitproj` is a complete portable project snapshot.

The exported report PDF is an analysis summary, not the CurveMole software manual.

14.8 Exporting one numeric file per spectrum

Choose **File > Export spectra and fit curves...** or press **Ctrl+Shift+E** when the result should be immediately replottable as one multi-column table per spectrum. This is separate from the structured analysis bundle. Choose any project spectra and any of these options:

- export the measured spectrum with its fitted background subtracted;
- omit masked rows;
- include every enabled individual model function;
- include the combined fitted background;

- include the total fit;
- include residuals as data minus total fit.

The x and spectrum columns are always written. Output names follow <spectrum>_curvemole.<original extension>; duplicate names receive a numeric suffix. The original delimiter is retained when practical, otherwise the output is tab-delimited. Headers describe axis labels, units, functions, background state, total fit, and residuals, and never contain whitespace so they remain safe for whitespace parsers. An unknown original extension is preserved.

15. Command-line interface

15.1 General form

```
curvemole [--version] [--json] [--verbose] COMMAND [OPTIONS]
```

Use `--json` for machine-readable output. Normal data go to standard output; logs and errors go to standard error.

15.2 Open the GUI

```
curvemole gui
```

The desktop executable can also receive a project or data path as its first argument.

15.3 List available functions

```
curvemole functions list
curvemole functions list --json
```

15.4 Fit one file

```
curvemole fit spectrum.csv \
  --x wavenumber \
  --y absorbance \
  --function gaussian \
  --background linear \
  --output spectrum.fitproj
```

`--x` and `--y` accept a column name or zero-based index. Optional initial values are `--center`, `--width`, and `--area`. The meaning of `--width` follows the selected function: Gaussian and Voigt use sigma, Lorentzian uses gamma, and pseudo-Voigt uses fwhm.

If no initial peak override is supplied, the CLI uses the strongest automatic positive peak suggestion when available. Background choices are none, constant, or linear.

15.5 Fit a series

```
curvemole fit-series scan_01.csv scan_02.csv scan_03.csv \
  --x 0 --y 1 \
  --function pseudo_voigt \
  --background constant \
  --mode sequential \
  --output series.fitproj
```

Modes are independent, sequential, and global. Use `--global-search` to request Differential Evolution, but remember that the simple fit CLI does not provide bound arguments. For bounded global searches, use a YAML workflow or the Python API.

15.6 Inspect and validate

```
curvemole inspect analysis.fitproj --json
curvemole inspect raw_data.dat --json
curvemole validate analysis.fitproj
curvemole validate workflow.yml
```

Inspecting a data file reports detected parsing and a preview. Validating a project checks archive structure and checksums. Validating YAML checks the supported workflow schema and required top-level structures.

15.7 Export an existing project

```
curvemole export analysis.fitproj exported_analysis
```

Use `--versioned` for a timestamped subdirectory. Use `--force` only when updating a directory whose CurveMole ownership manifest permits the overwrite.

15.8 Exit codes

Code	Meaning
0	Success
2	Command usage error
3	Data validation error
4	Fit error
5	Project format error
6	Plugin or other CurveMole domain error
10	Unexpected failure

16. Reproducible YAML workflows

16.1 Running a workflow

```
curvemole run examples/gaussian_workflow.yml --json
```

Relative paths are resolved from the workflow file's directory. Schema version 1 supports imports, custom functions, explicitly trusted plugins, models, constraints, fit settings, project output, and bundle export.

16.2 Complete example

```
schema_version: 1
name: Gaussian workflow example
series_name: Synthetic spectra

imports:
  - path: gaussian.csv
    config:
      delimiter: ","
      decimal: "."
      header: true
    columns:
      x: x
      y: [y]
      sigma_y: sigma_y
    aliases: [Synthetic Gaussian]

models:
  - curve: Synthetic Gaussian
    components:
      - function: constant
        name: Background
        parameters:
          offset: 0.0
      - function: gaussian
        name: Main band
        parameters:
          area: 3.0
          center: 0.3
          sigma: 1.0
        constraints:
          sigma:
            minimum: 0.001
            maximum: 5.0

fit:
  mode: independent
  curves: [Synthetic Gaussian]
  settings:
    solver: local
    loss: linear
```

```
seed: 1729

export:
  project: gaussian_result.fitproj
  bundle: gaussian_export
  versioned: false
  overwrite: false
```

16.3 Custom formulas in YAML

```
custom_functions:
- identifier: exponential_decay
  display_name: Exponential decay
  kind: generic
  formula: offset + amplitude * exp(-x / tau)
  defaults:
    offset: 0.0
    amplitude: 1.0
    tau: 1.0
  bounds:
    tau: [0.000001, 1000000.0]
  derived:
    lifetime: tau
```

Use the custom identifier in a model component exactly like a built-in function.

16.4 Trusted plugins

List local plugin manifest paths under plugins. Unattended execution also requires one `--trust-plugin IDENTIFIER` argument for each plugin whose Python source may run:

```
curvemole run workflow.yml \
  --trust-plugin org.example.lineshape
```

Trust is never inferred merely because a manifest is present.

17. Python API

17.1 Stable package-level objects

The package root exports GUI-independent classes including Curve, Series, Dataset, Component, Model, Parameter, Project, FitSettings, FitPlan, FitResult, Fitter, FunctionRegistry, and default_registry.

17.2 Minimal fit

```
import numpy as np

from curvemole import Component, Curve, FitSettings, Fitter, Model

x = np.linspace(-5.0, 5.0, 1001)
y = 0.1 + 3.0 * np.exp(-0.5 * ((x - 0.4) / 0.8) ** 2)

curve = Curve("Synthetic band", x, y)
model = Model("One Gaussian")
model.add(Component.create("constant", initial={"offset": 0.1}))
model.add(
    Component.create(
        "gaussian",
        initial={"area": 6.0, "center": 0.3, "sigma": 1.0},
    )
)

result = Fitter().fit_single(curve, model, FitSettings(loss="linear"))
print(result.success)
print(result.statistics)
```

The Gaussian expression used to generate y above is height-parameterized, while the CurveMole Gaussian component is area-parameterized. This is why the initial model area is not simply the synthetic height.

17.3 Bounds and fixed values

```
peak = model.components[1]
peak.parameters["center"].minimum = -1.0
peak.parameters["center"].maximum = 1.0
peak.parameters["sigma"].minimum = 0.05
peak.parameters["sigma"].maximum = 3.0
peak.parameters["area"].fixed = False

for parameter in peak.parameters.values():
    parameter.validate()
```

17.4 Cross-curve links

```
source_path = model_a.parameter_path(curve_a.id, peak_a.id, "center")
peak_b.parameters["center"].link = "${" + source_path + "} + 5.0"
```

Construct a Global FitPlan containing both curve IDs. CurveMole validates the complete graph before optimization.

17.5 Global fit

```
from curvemole.core.fitting import FitMode

plan = FitPlan(
    [curve_a.id, curve_b.id],
    mode=FitMode.GLOBAL,
    settings=FitSettings(loss="linear"),
    spectrum_weights={curve_a.id: 1.0, curve_b.id: 1.0},
    equal_contribution=True,
)

result = Fitter().fit(
    plan,
    {curve_a.id: curve_a, curve_b.id: curve_b},
    {curve_a.id: model_a, curve_b.id: model_b},
)
```

17.6 API compatibility during Preview

The package root is intentionally smaller and more stable than internal modules, but version 0.x may still introduce incompatible changes. Pin the exact CurveMole version in automated research workflows and record it with results.

18. Function Builder and plugins

18.1 Formula Builder

Open **Tools > Function Builder**. Enter:

- an identifier using letters, numbers, and underscores;
- a display name;
- classification as peak, background, or generic;
- a formula in x ;
- optional formulas for derived area and FWHM;
- for peak functions, optional semantic roles for detected parameters.

CurveMole detects every non-function symbol other than x as a parameter. Newly detected parameters default to 1.0 and are unbounded. After adding the function to the project library, add a component and edit its initial values and bounds in the normal parameter table.

Example:

```
offset + amplitude * exp(-x / tau)
```

Detected parameters are amplitude, offset, and tau.

For a function classified as a peak, the **Peak parameter roles** table can assign at most one of each supported meaning:

- peak position / centre;
- peak height or peak area / integral;
- FWHM, Gaussian sigma, or HWHM/gamma.

These roles tell graphical placement and automatic peak detection how to initialize parameters even when their names are arbitrary. Height and area roles are mutually exclusive. A width role also supplies the standard derived FWHM automatically when a separate FWHM formula was not entered. Leave genuinely unrelated parameters as **Not specified**. An explicitly unassigned role is respected; CurveMole does not guess a meaning from its name.

18.2 Custom function persistence

The first time a formula is added, CurveMole asks the user to create a dedicated `my_curvemole_functions` folder at a chosen location or open an existing folder with that exact name. Each definition is saved there as `<identifier>.curvemole-function.json`, registered in the current application, and also embedded in the project. Formula, classification, derived quantities, and peak roles travel together.

CurveMole remembers the selected folder and loads every valid function from it at each startup. Reusable functions therefore appear in the normal Add dialog, Quick Add selector, and automatic peak-shape selector without requiring a particular project to be open. Copy the folder between computers to move a personal function library; open that folder when CurveMole asks for the library location.

Project-contained definitions remain portable. When a project reopens, CurveMole reconstructs and validates each restricted expression. A malformed user function or library file is skipped and recorded in the Log rather than executed. Formula files use the safe expression system and are distinct from executable Python plugins.

18.3 Local plugin structure

A local plugin consists of a manifest ending in `.curvemole-plugin.json` and a Python module in the same directory. Example manifest:

```
{
  "identifier": "org.example.my-lineshape",
  "version": "1.0.0",
  "api_compatibility": "1",
  "licence": "GPL-3.0-or-later",
  "capabilities": ["functions"],
  "module": "my_lineshape.py"
}
```

The module must expose `register(registry)` and may register `FunctionDefinition` objects. Full details are in [plugins.md](#).

18.4 Trust boundary

Formula Builder expressions are interpreted safely. Plugins are different: a plugin is Python code and can perform any action allowed to the current user account.

CurveMole reads manifests before executing a module and requires an explicit trust decision. Review source, origin, license, and version before trusting. Do not trust a plugin merely because its identifier resembles a known package.

18.5 Plugin Manager

Open **Tools > Plugin Manager**, choose a directory, and scan. Select a candidate to review its metadata, then explicitly trust and load it. Trusted identifiers and the plugin directory are stored with project UI state. Trust should be reconsidered when plugin source or version changes.

19. Troubleshooting

19.1 The application does not start on Linux

Run the AppImage from a terminal to capture the exact error. If Qt reports missing EGL, OpenGL, XKB, or XCB libraries, install the distribution packages listed in Section 3.3. If FUSE mounting fails, use `APPIMAGE_EXTRACT_AND_RUN=1`.

The AppImage is built on Ubuntu 22.04 for broad glibc compatibility. Very old Linux distributions may still require a newer runtime.

19.2 Quick Guide, manual, updates, or issue links do not open

CurveMole attempts to launch the host operating system's opener without exposing the AppImage's private Qt and C++ libraries to it. If the opener still fails, the dialog shows the path or URL to copy manually.

Use these direct addresses:

```
https://github.com/SebRoLENS/curvemole
https://github.com/SebRoLENS/curvemole/releases
https://github.com/SebRoLENS/curvemole/issues
```

19.3 Windows SmartScreen or macOS Gatekeeper warns

The Preview executables are not yet commercially signed or notarized. Confirm that the file came from the official release, verify its SHA-256 checksum, and use the operating system's documented override only if you accept the risk.

19.4 Import produces one column

The selected delimiter is probably wrong. Reopen import and compare whitespace, comma, semicolon, tab, or pipe. A decimal comma file commonly uses semicolon as its delimiter. Confirm the header choice and preview before mapping.

19.5 Imported points disappear from fitting

Inspect the Worksheet. A point is excluded when x or y is not finite, σ_y is invalid, a weight is invalid, or the point is masked. Faded markers indicate masks. Use Unmask or undo the transformation that generated missing values.

19.6 A peak appears at the wrong height after placement

Built-in peaks use integrated area, not height. The graphical initializer estimates height relative to the existing model and converts it to area. For overlapping peaks or a poor background, adjust the center target, width handle, and area value before fitting.

19.7 A spline looks unstable

Use fewer, well-spaced nodes. Place nodes where background information exists rather than at every data feature. Remember that every node y value is a fit parameter by default. Fix or bound nodes when the data do not independently determine them.

19.8 Differential Evolution requests finite bounds

Set a finite lower and upper bound for every free parameter. A linked parameter does not need an independent search bound, but its resulting value must satisfy its own bounds. If defensible bounds are unavailable, improve initial estimates and use the local solver.

19.9 The fit is underdetermined or covariance is unavailable

Reduce the number of free parameters, fix known values, add justified links, widen the data range, or improve signal information. More optimizer evaluations cannot create information absent from the data.

19.10 A parameter remains at a bound

First verify the initial value and physical bound. Then inspect correlations and profile likelihood. A bound-active estimate often has an asymmetric uncertainty and may indicate that the data do not locate the optimum within the allowed region.

19.11 A sequential fit pauses

The affected curve becomes active and the status bar displays **Continue sequential fit**. Inspect its propagated model, initial/fitted values, masks, and constraints. The Log and notification identify whether fitting failed, normalized residuals worsened, or a monitored parameter changed beyond its threshold. Correct or approve the model, then continue; this spectrum becomes the new source. For a series with legitimate abrupt changes, adjust or disable the relevant safeguard, or exclude a known discontinuous function from only the parameter-change monitor.

19.12 A project opens read-only

Another instance may hold the sibling .lock file. Close the other instance, or use Save As. Remove a lock manually only after confirming it is stale.

19.13 Export refuses to overwrite

Choose a new directory or a versioned export. If updating an existing bundle, keep its .curvemole-export.json manifest and explicitly enable update. CurveMole will not claim ownership of unrelated colliding files.

19.14 A resampling analysis has many failed replicates

The baseline may be near bounds, poorly initialized, or structurally unstable under perturbation. Increase robustness of the model before increasing replicate count. Report completed and failed counts with any interval.

19.15 An update is detected but cannot be installed automatically

Automatic replacement is limited to a running Linux x86-64 AppImage or Windows x86-64 standalone executable whose directory is writable. Finish or cancel active tasks. Python/source installations, macOS packages, unsupported architectures, and protected application directories must be updated from the GitHub release page. On Windows, a failed staged update records a helper log in the system temporary directory; include that log when reporting a reproducible updater failure.

19.16 Reporting a problem

Use [GitHub Issues](#). Include:

- CurveMole version;
- operating system and package type;
- exact steps to reproduce;
- expected and observed behavior;
- terminal output or traceback;
- a minimal non-confidential data file if possible.

Review every attachment. Diagnostic material must not contain unpublished spectra, personal information, credentials, or proprietary data unless you intentionally make them public.

20. Reproducible research, privacy, and citation

20.1 Recommended research record

Archive at least:

- exact CurveMole version and DOI;
- original data and import mapping;
- complete `.fitproj`;
- analysis bundle;
- model functions and component order;
- initial values, bounds, fixed values, and links;
- masks and transformations;
- point uncertainties or weights;
- fit mode, spectrum weights, solver, and loss;
- random seed and replicate counts for uncertainty analyses;
- warnings, failed replicates, and residual diagnostics.

20.2 Local processing and privacy

CurveMole performs fitting locally. It has no telemetry and does not upload data, projects, models, results, or formulas. Its automatic startup and hourly version check requests only the public latest-release metadata from GitHub. Opening documentation or release notes, downloading an update, reporting an issue, or using Zenodo is otherwise an explicit external action.

Plugins run locally with the user's permissions and can violate these expectations; trust only reviewed plugin code.

20.3 Citation

If CurveMole contributes to published work, cite the exact version used. The release DOI is inserted into `CITATION.cff` and the repository README after Zenodo archival. Until archival completes, the versioned GitHub release is the authoritative record:

Romi, S. (2026). *CurveMole: Modular Scientific Curve Fitting* (Version 0.19.2)
[Computer software]. GitHub. <https://github.com/SebRoLENS/curvemole/releases/tag/v0.19.2>

The repository provides **Cite this repository** from `CITATION.cff`.

20.4 Author and contact

Sebastiano Romi
European Laboratory for Non-Linear Spectroscopy (LENS)
University of Florence (UNIFI)
romi@lens.unifi.it

20.5 License

CurveMole is released under GPL-3.0-or-later. User data, private projects, results, and unpublished formulas remain under the user's control. Publicly distributed extensions must follow their applicable license obligations.

Appendix A. Keyboard shortcuts

Shortcuts use the platform's standard key sequence where applicable.

Shortcut	Action
Ctrl+N	New project
Ctrl+O	Open project
Ctrl+I	Import data
Ctrl+S	Save project
Ctrl+Shift+S	Save project as on common desktop mappings
Ctrl+E	Export analysis bundle
Ctrl+Shift+E	Export one numeric file per selected spectrum
Ctrl+Z	Undo
Standard Redo shortcut	Redo
Ctrl++	Add component
F5	Open Fit dialog
F1 or platform Help shortcut	Quick Start
Esc	Cancel graphical peak or spline placement
Ctrl while dragging	Change a fixed graphical parameter without unfixing it
Right-drag on plot	Mask or unmask an interval directly
Up / Down with plot focus	Activate previous / next spectrum

Appendix B. Built-in identifiers

Identifier	Display name	Parameters
gaussian	Gaussian	area, center, sigma
lorentzian	Lorentzian	area, center, gamma
voigt	Voigt	area, center, sigma, gamma
pseudo_voigt	Pseudo-Voigt	area, center, fwhm, eta
constant	Constant background	offset
linear	Linear background	intercept, slope
polynomial	Polynomial background	c0 through selected order
cubic_spline	Cubic-spline background	y0 through last node

Appendix C. File extensions

Extension	Purpose
.fitproj	Complete CurveMole project
.fitmodel	Reusable model and formula definitions
.curvemole-function.json	Reusable safe formula in my_curvemole_functions
.curvemole-plugin.json	Manifest for an executable Python plugin
.yml, .yaml	Reproducible workflow
Any text-file suffix	Imported data when the contents form a valid numeric table
.whl	Installable Python package
.AppImage	Linux desktop package
.exe	Windows desktop executable
.dmg	macOS disk image

Appendix D. Preview 0.19.2 limitations

The following boundaries are important when evaluating this release:

- scientific validation toward 1.0.0 is still in progress;
- only nonlinear least squares is implemented as the final local optimizer;
- `sigma_x` is stored but not used in optimization;
- the GUI does not yet provide a dedicated parameter-path picker for complex links;
- fit ranges exist in the core model but do not yet have a complete graphical editor;
- the GUI does not yet provide one consolidated table for every fit statistic;
- autosave recovery exists, but there is no automatic startup recovery chooser;
- Windows and macOS packages may be unsigned; Linux release provenance is available through GitHub artifact attestation;
- the interface and manual are currently maintained in English;
- plugin API and other 0.x APIs may change before 1.0.0.

These limitations are stated so that testing can focus on genuine behavior rather than inferred promises. Report unclear workflows and reproducible defects through the issue tracker.

Appendix E. Building this manual

The Markdown file you are reading is the only hand-edited manual source. From a source checkout with Pandoc, XeLaTeX, and latexmk installed, run:

```
python scripts/build_manual.py
```

The command validates version declarations and local links, then creates:

```
docs/CurveMole_User_Manual.tex  
docs/CurveMole_User_Manual.pdf
```

GitHub Actions repeats this process for documentation changes and releases. Release assets use versioned filenames and include both LaTeX and PDF editions. Do not edit the generated files directly; changes will be replaced by the next automated build.